Activity 4.6 – Pipeline Statistics

1.Design Approach:

Designed a pipeline performance statistics module using Verilog HDL to monitor total clock cycles, retired instructions, stall cycles and flush cycles. Implemented independent counters for each parameter.

2.Verification Steps:

- Created a separate testbench.
- Applied reset.
- Generated retirement signals.
- Activated stall conditions.
- Activated flush conditions.
- Verified all counter outputs.
- Simulated using ModelSim.

3.Results:

The testbench checks cycle, retirement, stall and flush counters. PASS/FAIL messages are displayed.

4.Issues Faced:

Each counter must increment only under its respective condition. Simultaneous events must be handled correctly. Reset must clear all counters.

5.Conclusion:

Successfully designed and verified a pipeline statistics monitoring module using Verilog HDL.